Huawei Access Node Optical Transport Failure SOP (KOR-ACC-83)

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_1332

Description

The Optical Transport Failure in the Access Node KOR-ACC-83 at Koramangala Datacenter is primarily caused by degradation in transceivers (SFP/XFP) or fiber cuts resulting in received (Rx) and transmitted (Tx) power loss. This failure domain occurs due to the Huawei Access Node's role in handling optical transport links, which are critical for data transmission between nodes. The protocol mechanics involve the use of Optical Channel Multiplexing (OCM) and Optical Channel Transport Network (OCTN) to manage the transport of optical signals. However, physical degradation of transceivers or fiber cuts compromise the integrity of these signals, leading to power loss. This results in the ALARM_1332 warning being triggered, indicating a potential issue with the optical transport link.

Pre-requisite Checks

- Verify that the Koramangala Datacenter is accessible and that remote hands are dispatched to the site.

- Check the status of redundant link connections to ensure they are available and functional.

- Perform a site survey to identify any potential physical damage to the fiber or transceivers.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP Transceivers	High
Fiber Cuts or Physical Damage	Medium
Optical Signal Attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-channel link info, Expected Output: Link status: Up, Tx power: 0 dBm, Rx power: -20 dBm, Description: This command is used to display the status of the optical channel link, including the link status, transmitted (Tx) power, and received (Rx) power. The expected output indicates that the link is up, but the received power is significantly lower than the transmitted power, suggesting a potential issue with signal attenuation or power loss.

Command: display optical-channel alarm info, Expected Output: Alarm status: ALARM_1332, Description: Optical transport link failure, Description: This command displays the alarm status for the optical channel, including the alarm code and description. The expected output indicates that the ALARM_1332 has been triggered due to an optical transport link failure.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP Transceivers, Command: replace optical-channel transceiver 1/1/1, Description: This step involves replacing the degraded SFP/XFP transceivers with new ones to restore the optical transport link. The command replaces the transceiver on slot 1 of the optical channel.

Step 2: Step: Repair or Replace Fiber, Command: replace optical-channel fiber 1/1/1, Description: If the issue is due to fiber cuts or physical damage, this step involves repairing or replacing the damaged fiber to restore the optical transport link. The command replaces the fiber on slot 1 of the optical channel.

Post-Check Verification

```
Command: display optical-channel link info, Expected Output: Link status: Up, Tx power: 0 dBm, Rx power: -10 dBm, Description: This command is used to verify that the optical channel link is functioning properly after the resolution steps. The expected output indicates that the link is up, and the received power is within acceptable limits.

Command: display optical-channel alarm info, Expected Output: Alarm status: No alarm, Description: Normal operation, Description: This command verifies that the ALARM_1332 has been cleared and the optical transport link is functioning normally. The expected output indicates that there are no alarms, and the system is operating within normal parameters.
```

Roll-back Procedures

- Step: Restore Original Transceivers or Fiber, Command: replace optical-channel transceiver 1/1/1 original, Description: If the fix fails, this step involves restoring the original transceivers or fiber to ensure the system returns to its previous state.

- Step: Verify System Integrity, Command: display system status, Description: This step involves verifying that the system is functioning properly after the rollback procedure. The command displays the system status, which should indicate that the system is operating normally.